



Computer-Integrated Manufacturing

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GLOSSARY

bill of material (BOM) A listing of all the raw materials, parts, subassemblies, and assemblies needed to produce one unit of finished product. The listing in the BOM file is hierarchical and it shows the quantity of each item needed to complete one unit of the following level of assembly.

computer-aided design (CAD) This is defined as a computer-based system that provides interactive graphic facilities to assist engineering in product and tool planning, design, drafting, design revisions, process planning, facilities planning, and design optimization. CAD is also used to maintain BOM and engineering revision information and software system.

computer-aided manufacturing (CAM) A computer-based system to program, direct, and control manufacturing process and handling equipment. These programs are based on the geometry of the parts and tool paths, both of which are captured by the CAD system. CAM serves as the primary automated communication link between the engineering and shop floor functions of a company, thus integrating design and manufacturing activities.

computer-integrated manufacturing (CIM) This is viewed as an integrated information system that links a broad range of manufacturing activities, including engineering design, flexible manufacturing systems, production planning and control, and all the business functions in any production/operation organization. The term was first used by Har-

rington in 1973 to envision the integration of computerized manufacturing systems at the shop floor level and management information systems at the corporate level. Since then, the concept has been expanded, both technologically and philosophically, to integrate information technologies into the entire organizational functions and units, including engineering, manufacturing, and business functions as well as all managerial level and operational functions.

enterprise resource planning (ERP) The ERP system, adopted in recent years by large and medium-size firms, is defined as strategic business solution that integrates all the business functions, including manufacturing, finance, and distribution. ERP systems encompass traditional transaction processing systems as well as model-based DSS such as data warehouse, supply chain optimization, planning, and scheduling systems. Such integrated systems improve management of information resources and enable decisionmakers to better access required information across the organization. Software vendors such as SAP AG, Baan, PeopleSoft, and Oracle provide a host of integrated ERP products.

flexible manufacturing system (FMS) This a group of machines designed to handle intermittent processing requirements and produce a variety of similar products. The system includes supervisory computer control, automatic material handling, and robots or other automated processing equipments.

manufacturing resource planning (MRP II) This is an expanded scope of MRP and a broader approach

to production resource planning that involves not only those in MRP but also other areas, such as marketing and finance, of the firm in the planning process. A major purpose of MRP II is to integrate primary functions and other functions such as personnel, engineering, and purchasing in the planning process. The MRP II systems have also the capability of performing simulation, enabling managers to answer a variety of "what if" questions so that they can gain a better appreciation of available options and their consequences.

master production schedule (MPS) This is a timetable that specifies what to be made and when. The schedule must be in accordance with a production plan that sets the overall level of output in broad terms (e.g., product families, standard hours, or dollar volumes). The plan also includes a variety of inputs, including financial plans, customer demand, engineering capabilities, labor availability, inventory fluctuations, supplier performance, and other considerations.

material requirement planning (MRP) This is an information system designed to handle ordering and scheduling of dependent-demand inventories, e.g., raw materials, component parts, and subassemblies. A production plan for a specific number of finished products is translated into requirements for component parts and raw materials working backward from the due date, using lead times and other information to determine when and how much to order. Hence, requirements for end items generate requirements for low-level components, which are broken down by planning periods (e.g., weeks) so that ordering, fabrication, and assembly can be scheduled for timely completion of end items while inventory levels are kept reasonably low.

model-based management system (MBMS) A model-based management system makes use of quantitative models (i.e., Operations Research and Management Science techniques) to assist the decisionmaker in improving effectiveness of his/her decision processes.

transmission control protocol–internet protocol (TCP–IP) Two elements of a suite of internet protocols which include transport and application protocols. The combined acronym TCP–IP is commonly used for the whole suite. While the TCP element divides a message up into packets (call datagrams), recombining the message at the receiver, the IP element routes the package across the network.

I. INTRODUCTION

The focus of Information Technology (IT) applications has shifted dramatically over the past two decades from efficiency/process improvement to strategic/competitive advantages. This change of focus has inspired organizations to take a business vision, rather than a technological focus, toward IT deployment and management. While computer-integrated manufacturing (CIM) has been recognized as one of the manifestations of IT evolution with much potential for strategic opportunities and competitive advantages, the traditional approach in management of technology, particularly focusing on manufacturing and engineering aspects of the system, in many corporations has remained as an obstacle to the successful implementation and management of CIM.

In practice, CIM has emerged as a new framework in the evolutionary process of development of new approaches and techniques in production/operations management such as just-in-time (JIT), total quality management (TQM), flexible manufacturing systems (FMS), computer-aided design (CAD), computer-aided manufacturing (CAM), material requirement planning (MRP), and manufacturing resource planning (MRP II), among others. In this context, CIM provides a framework to integrate a variety of technologies in support of information requirements of the company's engineering, manufacturing, business, and management functions. This integration facilitates data communication and data sharing among various functional areas within an organization and thus supports decision-making processes, reduces lead time, and improves communication among decision makers within and outside of the organization.

Since the early 1970s, computers have been increasingly utilized in various operations of the production floor. CAD has extended design and drawing capabilities; MRP has reduced lead time, cut waste, and reduced down time; and software to support Master Production Scheduling (MPS) has increased efficiency and cut production costs. Production floor management of automated machines has been integrated with CAM to form a direct line between the drawings as conceived by the design engineer and the actual production of parts and products. Computer-aided engineering (CAE) has been integrated with CAD to allow an immediate analysis of designs as they take shape with CAD graphics. JIT inventory control, as a stockless or zero-inventory concept, is an integration of production schedules, warehouse management data, and material handling data, both within and between com-

pany files and those of subcontractors. While these advances have resulted in significant short- and long-term benefits, however, information technology in all these situations has been developed and adopted to achieve functional efficiency and measurable benefits. In many cases the sole objective of automation has been toward downsizing. Because of technological factors as well as organizational infrastructure, the managerial attitude toward exploring these technologies has been an engineering approach. Thus, making it often functional or departmental rather than company-wide strategy. The result has been "islands of automation" in which individual processes have been automated without much concern for integration and compatibility among them. These processes have had limited flexibility because of their lack of integration and the limited information-sharing capability.

The purpose of this study is to provide an overview of CIM and its component technologies, in a conceptual and managerial rather than a technical context, and to analyze the organizational environment and managerial approaches required for successful implementation and applications of CIM. In particular, CIM is viewed as the integration of computerized manufacturing systems at the shop floor (CMS) and Management Information Systems (MIS). This approach implies an integrated, company-wide, and MIS approach toward CIM implementation and management rather than traditionally functional and technological attitude. This study, therefore, emphasizes the key role of senior management and top MIS professionals in planning, selecting, justifying, implementing, and managing a CIM system. Responses to a survey targeted at various professionals and personnel involved in managing and using CIM will be used to identify and analyze some managerial perceptions that are critical to successful planning and implementation of CIM.

In Section II, we provide various views on the concept of CIM. Next, we present a conceptual schema of a CIM system and its major components. It is our contention that CIM should provide information to different organizational units. This must be achieved through well-integrated database and model-base systems, such as described in Section IV. In Section V, we elaborate on management issues and strategic dimension CIM initiatives. An integrated CIM system requires champions from all organizational units. However, recent literature reflects CIM as an engineering initiative, and, thus avoiding input from senior executives for linking CIM to the organizational strategy. Obviously, if this is found to be true, then organizations do not realize the full potential of CIM. Therefore, this

conjecture was the basis of an empirical investigation that we conducted. The nature of this study and findings is reported in Section VI. Section VII provides a summary and concluding comments.

II. CONCEPTION OF CIM AND LITERATURE REVIEW

Over the past two decades, information technology has significantly changed the way organizations compete, and has also changed the structure of the entire industries. Organizations that have used their business vision, experience, and their expertise to effectively integrate IT into their corporate strategy and organizational plans have gained much competitive advantage and success in the marketplace. In this context, CIM has surfaced as a concept that integrates information technologies into the entire organizational functions and units, including engineering, manufacturing, and business functions as well as all managerial level and operational functions. Since its first use in the term CIM has increasingly received a great deal of attention from production and inventory control managers, consultants, and scholars. Along with a JIT approach in manufacturing, TQM, and world class manufacturing, CIM has increasingly represented an important path that manufacturing companies could follow to improve their competitiveness in domestic and global markets.

CIM has been viewed and interpreted differently in the literature. There has been a tendency in the past to consider CIM as a purely technological endeavor in meeting short-term goals and as a quick payback solution to manufacturing problems. In this context, CIM has been viewed as the most advanced manufacturing technology known, or as the use of database and communication technologies to integrate the design, manufacturing, and business functions that comprise the automated segment of the facilities. It has also been defined as a complete application of computer-aided technology, and as the integration of CAD/CAM and production management (encompassing all activities, from planning and design of a product through its manufacturing and shipping. CIM has been viewed as a global approach which aims at improving industrial performance. This approach is applied in an integrated way to all activities from designing to delivery and after sale. The objectives of the CIM in this approach are to simultaneously improve productivity, decrease costs, meet due dates, increase product quality, and secure flexibility at local

and global levels in a manufacturing system. In such an approach, economic, social, and human aspects are at least as important as technical aspects. On the one hand, CIM is viewed as almost synonymous with the use of machine vision, automated handling systems, robotics, and flexible manufacturing systems. On the other hand, CIM is primarily considered an information-based system to manage and structure databases. However, recent trends in the literature acknowledge CIM to be primarily a strategic system that draws on IT to help the firm better meet the needs of its market place. Thus, much of the rapidly proliferating literature on CIM has dwelt on its technological aspects and its potential.

Industry-specific definitions of CIM also are prevalent in the literature. The print industry, for example has adopted certain CIM principles such as electronic data interchange (EDI), digital linkage of users and suppliers, and automation in the production and binding steps to move the industry from a labor-intensive position to one of greater automation. The machine tool industry is another sector frequently cited as practicing certain CIM concepts such as CAD, computer-aided process planning (CAPP), CAM, and MRP II. For industry-specific definitions, it appears that CIM provides a vast menu of possibilities for an organization. From this list of possibilities, decisionmakers select various processes, hardware and software which are then adapted to specific industry needs.

Another emerging trend in CIM literature involves Product Data Management (PDM) systems. In today's manufacturing environments, the need for PDM is paramount because of rapidly changing product technology. In addition to PDM, the DBMS (for example, an Oracle system) must be capable of integrating with a particular network protocol (e.g., TCP-IP). This integration is essential for the development, delivery, and support of object-oriented client/server documents that are critical to business functions. For successful CIM installation, PDM cannot be overlooked because it is the backbone of data management in a CIM environment.

In practice, there is a wide array of CIM systems implemented at varying levels of sophistication with varying degrees of success. Particularly, in gaining considerable increases in productivity, reduction in lead time, and unit cost as well as in better machine utilization. For instance, GM redesigned its full-size pickup track operations at the Pontiac East Assembly Plant, Michigan, its largest manufacturing automation protocol (MAP) installation worldwide and used CIM to reduce redundancy and improve quality and customer service. This system handles almost the en-

tire operation (99%) of truck cab and box welds, application of vehicle base paint by robots, and inspects truck boxes and doors by vision systems to ensure the accuracy of parts going through the welding process. Over 300 programmable controllers direct robots to different weld times to regulate the time and the pressure of welds.

Ford Motor Company plastic plant, which manufactures 10,000 bumper fascias a day, installed its CIM in 1986, expanding since then. The automated system coordinates batch runs of numerous styles and colors of fascias through the plant. After molding parts are placed by robots onto automated guided vehicles (AGVS), which bring the parts over to an automated storage and retrieval system (AS/RS). When needed to fill an order, fascias are picked out of the automated warehouse and placed on an automated electronic monorail (AEM) conveyor, which carries the parts to assembly and painting operations. Once a customer places an order, the system automatically uploads the customer order into Milan's scheduling system, which will alert the AS/RS to find the parts and deliver them to assembly and painting station. One software package provides a graphical overview of the entire (AEM) conveyors. It alerts the operator to a fault in the track, and switches and dispatches maintenance to fix problems. A second software package does the same for AGVS.

GW Plastics in Bethel, VT, one of the CIM pioneers and CIM award winners, has seen its custom injection molding business soar by about 20% a year since its first installation of CIM in 1986, and has greatly expanded it to embrace its capacity; including new molding plants in San Antonio, TX, and Tucson, AZ. As business grew, GW replaced its original mid-range computers with an IBM AS400 mini-mainframe for greater processing power and installed a wide-area network for managing data in support of financial and manufacturing information, engineering files, sales, shipping, backlog, and forecast data. GW's bar-code system is integrated into AS400 to provide real-time verification of shipments to customers who demand JIT delivery.

Much of the literature on CIM suggests considerable improvements in productivity, lead time, and unit costs. However, the same studies also suggest that CIM systems appear not to have fully paid off in enhancing company-wide integration (technologically and organizationally), responsiveness, and business flexibility, as well as in supporting strategic decision-making processes. In some instances, the planning and implementation issues have led to disappointing results. Many firms do not achieve their objectives

from CIM investments largely because the analytical methods used to justify these projects often do not capture the richness and underlying flexibility of new technologies. Swamidass (1994) has surveyed United States manufacturers and found that even though manufacturing technologies such as CAD, CAM, TQM, JIT, and computer numerical control (CNC) have been widely used at most plants, skilled use is relatively uncommon among them. Although factory automation seems to be critical to the survival of manufacturing firms, it is not clear from current MIS or operations management literature whether CIM has been used mainly as a ubiquitous support function or as an important element in achieving a sustainable competitive advantage in products or markets. CIM as a technology is giving way to the idea of CIM as a philosophy. This includes "the highest level of integrated automation in a manufacturing plant" and increasingly broader conceptions of enterprise management. According to Forrester and Hassard (1992), part of the problem is that what counts as the effective integration of systems can be company specific. Hill (1994) contends that the unwillingness of firms to undergo the necessary organizational changes is one reason for their inability to gain the full benefits of CIM. Organizational integration also has to match and support the technological integration in the move toward CIM. De Meyer (1990) investigated the computerized integration of several business functions, such as inventory planning, accounting, and product design in European firms. In a series of cross-sectional studies, he shows that the most often integrated func-

tions include sales planning with master production scheduling and process control with quality reporting. His study does not address the technological impediments to integration or the relative contributions of the various enabling technologies to strategic management. Over time the decline in the cost of automation technologies has enabled manufacturers to adopt sophisticated production systems. Initial adoption of these systems has forced competitors to follow the industry leaders in the implementation effort or exit their industries.

III. CONCEPTUAL SCHEMA OF CIM AND ITS MAJOR COMPONENTS

Conceptually, CIM can be viewed as an integrated system shown in Fig. 1. Applications of CIM are in support of effective decision making processes in various functional areas, such as engineering, manufacturing, marketing, and accounting. In this holistic view, functional areas are considered as subsystems and evaluated against the company's goals and strategic objectives, namely, optimization of total company-wide business, rather than suboptimization. CIM is an interactive system in support of various types of decision problems (i.e., unstructured, semi-structured, and structured decision problems).

From technical standpoint, CIM requires integration of information through common databases, hardware/software integration, and compatible software. In particular, model bases and databases are

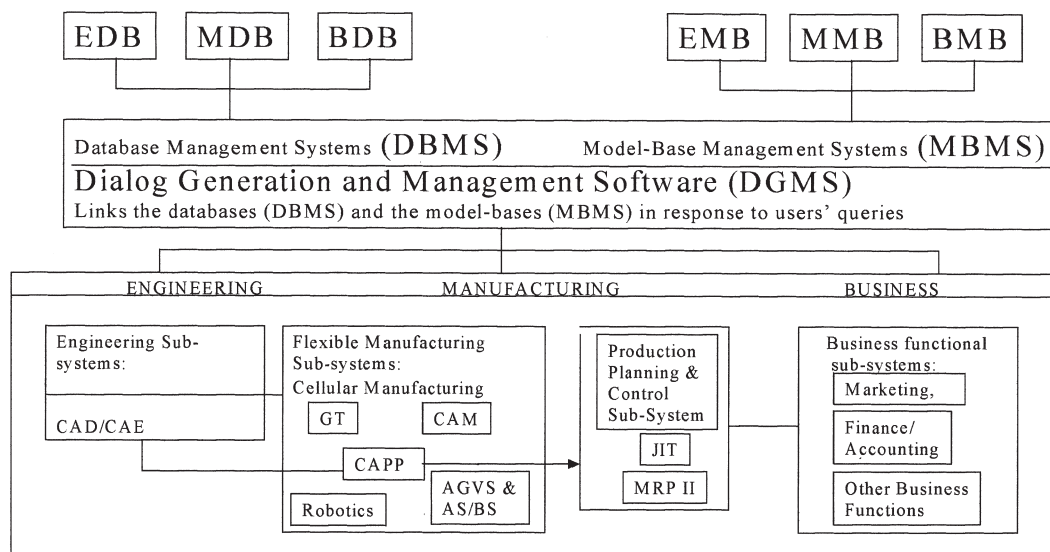


Figure 1 A conceptual schema of CIM.

linked through model based management systems (MBMS) and database management systems (DBMS). These software modules coordinate the flow of information manipulation and transmission in response to queries by users in various functional areas through dialog generation and management software (DGMS). The model-base component consists of interrelated and integrated models based on methodologies gleaned from engineering, operations research, and management science in support of decisionmakers from various functional areas. The abstraction of the models and their linkages are constructed from numerous variables common to various functional areas at different levels of aggregation. Based on a large number of internal linkages in the model-base and the database, once the optimum design is determined, manufacturing process would define the BOM. Other subsequent functional models are deployed to support the production planning and financial and accounting analysis.

Fixture planning and programming is an important function of CAPP, which is the link between design and manufacturing (Fig. 1). This generates instructions for the production parts and translates design data into process plan to produce a part or a product. CAPP has several layers which must be integrated with other functions within the process planning and other CIM activities. The CAPP subsystem must provide information and programs for machine tools, robots, and fixturing subsystems to the production department. Furthermore, the production planning and control (PPC) subsystem requires information about the resources necessary to manufacture the product. This information may include the required machine tools, fixturing subsystems, and process time, among others, which would allow PPC to plan and schedule the required resources.

Another technical challenge in CIM development is to manage the huge amount of rapidly changing heterogeneous data, which may range from graphic and geometric data in engineering to textual and numeric data in manufacturing, sales, financial, accounting, and other business areas. These various types of data are to be integrated and shared through an underlying integrated database system.

IV. INTEGRATION OF DATABASES AS THE CORE COMPONENT

Integration is the technical hub of CIM. One of the major requirements for a manufacturing system is the ability to store, manipulate, transmit, and display

graphical data integrated with text. For example, CAD/CAM must be capable of supporting engineering analysis with heavy loads of numerical computation, as well as manufacturing and business analysis. For instance, CNC machines, as the advanced manufacturing technologies of CIM, provide the data communication networks and thus define the level of data integration in the system. Manufacturing automation protocol and standard for the exchange of product data (STEP) provide common specifications hardware integration and data formats to facilitate communication between heterogeneous systems. Data from different subsystems such as CAD and CAM are shared through an underlying CIM database system, in a coordinated and adaptable format. Islam (1997) presented an analysis of IDEF, an enterprise modeling technique and data flow diagram (DFD), to identify some of the requirements for CIM database systems. He has identified some of the main characteristics of data model to support a generic CIM. In particular, the data from various subsystems like engineering, manufacturing, marketing, and other functional areas are different in nature—accounting data and other business systems are numeric and alphanumeric, while the data for CAD subsystems are geometric and graphic. Therefore, a common database should integrate the heterogeneous data type. Since design data have to be shared by subsystems like CAPP, MRP II, CAM, and FMS, the data need to be converted into a neutral recognizable format. The common database requires integrity constraints, such as a consistency checking routine that checks the validity between two design elements for their assembly. The CAPP subsystem generates a process plan based on the geometric and engineering data of the CAD subsystem as well as data from MRP II subsystems. The CAD data for BOM are part input data for MRP II subsystems, where geometric data are converted to alphanumeric data. The MRP II provides work center data, and purchase order and material plan which are used by other subsystems. The output from MRP II includes various data such as manufacturing order and inventory status, which are inputs to a CAM subsystem. CAM also uses design data from CAD and routine from CAPP subsystems. The FMS subsystem is a combination of heterogeneous hardware and software monitoring the manufacturing operations and thus should have updated information for CAM, the production plan, and other necessary changes in the system. Another issue in this context is the integration of the product data management (PDM) and enterprise resource planning (ERP). Companies often struggle in determining how these two technologies,

PDM and ERP, can best work together, what tasks each should handle, who should control information, and how they should be linked.

PDM evolved from engineering efforts in the 1980s and was first used primarily to manage engineering drawings and related CAD files (Miller, 1999). The technology eventually expanded to include management of data from a variety of separate applications including not only mechanical drafting, solid modelers, and structural analysis, but also electronic CAD (ECAD), NC programming, and others such as technical publications and office applications. In this capacity, PDM systems manage a variety of engineering data and processes including design geometry, project plans, part files, assembly diagrams, analysis results, correspondence, BOMs, specifications, engineering changes, approval processes, product structure, parts classification and retrieval, configuration management, program management, authorizations, workflow, and others.

Enterprise resource planning systems trace their roots back to manufacturing initiatives in the 1960s, evolving from MRP systems for inventory planning and control and later MRP II technology which expanded into shop floor scheduling and coordination. These systems broadened into what is now termed ERP, which encompasses numerous aspects of production such as inventory control, shop scheduling, capacity planning, and master scheduling as well as purchasing, sales, accounting, finance, and even human resources. By coordinating production operations for peak efficiency, ERP has become virtually indispensable for manufacturers to reduce manufacturing lead time and cost as well as to facilitate teamwork and collaboration on the factory floor. As PDM and ERP have both expanded, their increased functionality has resulted in overlap of some functions such as BOMs, parts classification, component information, configuration management, process workflow, and program management.

Transferring information quickly and accurately between engineering and manufacturing speeds workflow, improves communication throughout the organization, and avoids redundant efforts along with reducing associated errors and delays in recreating data. This provides an ability to leverage efforts so that data already entered in one area do not have to be recreated in another. In many cases, product information is transferred when design engineering releases a PDM product definition to industrial engineers, who then convert it into an assembly view for ERP. Often, a more direct exchange is being used where the PDM product structure is sent via a translator directly into ERP. A few systems even provide two-way translation

between PDM and ERP, giving engineering and manufacturing managers a direct view of one another's database. Still, in some other cases, companies strive for the most tightly integrated system where overlapping portions of PDM and ERP information are stored in the same database and shared by both systems.

Companies must identify a solution that can effectively integrate PDM and ERP systems and can closely meet their individual needs. Often, this means carefully evaluating company operations, goals, and procedures to identify how data such as BOMs can be shared, exchanged, and to ensure that the right information in the right format is available to the right person on a timely manner. As an example, fuel gauging and proximity sensing system manufacturer Eldec (a division of the Crane Co., Lynnwood, WA) masters all parts and BOMs in its PDM system and transfers information to its ERP system automatically at predefined times. Only six attributes are mastered in PDM, which establishes information such as purchase or fabricated part type and possible alternates. The data is sent to ERP, where an integration tool and template help create the 140 attributes needed for manufacturing. Production personnel fill in the remaining data and maintain control of information throughout manufacturing. To expedite the transfer of information from PDM to ERP, Eldec developed programs to automatically extract BOM data from schematics in PDM and put it into an Excel spreadsheet in a format that can be loaded directly into the ERP system.

ERP denotes control and management of the entire manufacturing facility in areas including not only production but also purchasing, finance, and engineering. It has evolved from earlier MRP systems for inventory control and later MRP II technology for shop floor scheduling and coordination. By coordinating the manufacturing operations for peak efficiency, ERP has become virtually indispensable for manufacturers to reduce manufacturing time and cost as well as facilitate teamwork and coordination. Similarly, PDM systems manage product-related information throughout the enterprise including design geometry, engineering drawings, project plans, part files, assembly diagrams, and product specifications. Typical users have traditionally included designers and engineers, but PDM system usage is being expanded to include other areas including manufacturing, sales, marketing, purchasing, shipping, and finance. Because of significant overlaps in data and functionality between PDM and ERP, most companies view integration between the two systems appealing.

The scope of integration in CIM is far beyond merely the hardware/software and data integration

and networking: the human and organizational aspects of CIM play a critical role in well-deliberated planning, successful implementation, and effective management of CIM.

V. STRATEGIC PLANNING, IMPLEMENTATION, AND MANAGEMENT ISSUES OF CIM

As we discussed earlier, CIM consists of overall integration of various functional areas from engineering to manufacturing, inventory, sales, marketing, etc., as well as integration of databases that support various levels of managerial decision problems. Since CIM requires such fundamental changes in organizational communication, both technologically and functionally, as well as massive capital investment, it is the senior management responsibility, with its long-range, company-wide view to integrate CIM strategy into the corporate strategy, to evaluate all possible strategies and intangible benefits, to judge its feasibility, and to decide on its adoption.

It has been reported that the failure of CIM to live up to its promise of a fully integrated information system has been due to a lack of commitment from senior management. While senior managers have taken full advantage of such innovative concepts as TQM and JIT scheduling, they have not fully utilized the economic benefits of CIM. Lack of understanding at the corporate level about technology management and its potential have been cited for the failure. More specifically, for nontechnical managers, it has been easier to envision the applications of the TQM and JIT concepts, and to measure their benefits. However, CIM has been perceived as a technological initiative that requires highly sophisticated hardware and software integration. Furthermore, the characterization of CIM technology has been depended on by the manufacturing industry and thus viewed as a highly technological issue rather than management issue.

Due to the technology orientation of CIM, it has been predominantly implemented by engineers and line managers. Consequently, the objective has been to improve efficiency and reduce costs rather than on improving corporate-wide integration of information systems. The engineering approach in development of CIM has also been implemented from bottom up where individual functional areas and lines have undertaken CIM projects based on their local needs and on a piecemeal approach, and thus has led to islands of automation, focusing on localized benefits. In particular, the lack of integration among functional areas such as marketing, purchasing, and production plan-

ning, and across key technologies such as MRP, CAD, and robotics, has been a stumbling block in the successful implementation and management of CIM processes. This is partly due to the lack of leadership from MIS professionals.

MIS professionals have traditionally focused on business functional areas such as accounting, finance, and planning data, and have shied away from robots and programmable controllers. Thus, MIS departments are lagging behind other departments in their contribution to CIM development.

To ameliorate this problem, senior management, particularly chief information officers, must approach CIM as a tool for competitive advantage in the market place, and integrate CIM strategy into corporate strategy and consider all possible strategic and intangible benefits as well as tangible ones. CIM technology contributes to competitive advantage by being responsive to the market changes and being flexible to redesign and manufacture according to market conditions, as well as in identifying the needs of a specialized market and responding effectively and efficiently to those needs.

Given the dimension and the characteristics of CIM strategy, it embraces considerations for the office and the business of the future as well as the factory of the future. The CIM strategy involves a dramatic change in manufacturing and business philosophy, as these changes will affect the entire corporation. Therefore, the need for a strategic plan detailing how a manufacturing and business concern can be addressed, via CIM, involves similar problems and issues inherent in information technology and MIS planning. The CIM strategy has to be viewed from an information technology management perspective and must be integrated into long-range business strategy by senior management. Senior managers must be involved in the CIM strategy development as the participants rather than spectators. Their roles should be shifted from being represented by subordinates to ones who create business strategy with CIM as an integral component. In this context, senior management can drive CIM strategy as an inseparable component of the overall business strategy.

VI. EMPIRICAL STUDY

The integration of various functional departments and managerial levels remains as a critical factor in design, development, implementation, and management of CIM systems. Managers, particularly those in small to mid-size companies, may assume CIM to rep-

resent a potential loss of management control, and a major challenge to their position and authority within the company. In their view, for instance, CIM may change the way the company plans, schedules, executes, and tracks production decisions. Much of the skills and knowledge possessed by the current management may no longer be pertinent to this new CIM-run environment; moreover, they may feel that they no longer understand the operations in CIM environment. Other employees may also be adversely disposed toward CIM. For example, they may view CIM aimed at narrowly automating much of the functions and reducing labor force rather than on developing knowledge workers and on improving the productivity and innovation. As a result, they may experience fear, uncertainty, and dread.

In order to examine the managerial issues that were previously discussed, a number of manufacturing companies in St. Joseph County, IN, were surveyed with regard to their adoption and implementation of CIM. The survey was carried out through a structured questionnaire format. A contact person in each company was identified for distribution of the forms among employees and collecting them after completion. An effort was made to include a wide variety of functional areas, types of positions, and managerial levels in the survey. The questionnaire included two different parts: (1) interviewer and organization identification: company size, position of interviewee, gender, the interviewers' level of education, and their perception of CIM impact on employment and on the stress to be placed on CIM initiatives; and (2) a set of open-ended questions. From a total of 211 respondents, 1.2% were employed in small companies (companies with less than 100 employees), 33.5% were employed in medium-size companies (with 100–500 employees), and 64.5% were from large companies (having more than 500 employees). The data in this survey thus represent mostly large and mid-size companies.

The survey consists of various employees: 23.9% of the respondents were from the management level, 51.2% professionals, and 24.9% clerical and other support personnel. With regard to the level of education, 12.2% of respondents had high school degrees or less, 13.8% had completed some college work, 22.8% had a bachelor's degree, 25.4% had carried out some graduate studies, 23.8% had completed a graduate degree, and 2.1% had professional training. Hence all educational levels except professional training are well represented in the survey. Various types of professionals were represented: 23.4% representing management of CIM, IS professionals made 23.9%, CIM professionals 27.9%, and clerical opera-

tors represented 24.9% of the respondents. Most companies that were visited during the data collection had computer systems of some sort. Often stand-alone accounting and inventory systems were in use, but these could not be easily interfaced with other manufacturing applications.

With regard to the computerization of the functional departments in their firms of those surveyed, 74.7% came from those with a fully computerized accounting department, 64.2% from firms with computerized MRP/MAP, 60.5% from those with computerized inventory, 54.7% from companies with computerized sales planning, 67.9% from computerized shop floor control, 76.3% from those with computerized process control, 71.1% from those with computerized quality reporting, and 16.3% from firms with other computerized departments. Thus, firms with computerized departments are well represented in the survey.

We have examined the perceptions of the respondents, through the analysis of variance (ANOVA), regarding their expected impact of CIM on the number of employees and their expected stress on CIM over the next two years. Table I reports the results of the statistical analysis when we focused on comparison of the perception of various groups of respondents. We examined the perceptions of various groups designated by the size of the companies, the gender of the respondents, the employment position, and more importantly the managerial positions of the respondents in the survey to the impact of CIM on employees and the stress to be placed on CIM initiatives.

We have used one-way ANOVA to analyze the perceptions of various respondents on the impact of CIM on employment and the level of stress to be placed on CIM initiative and management during the next two years. More specifically, we have examined the respondents' perceptions by a number of factors: the respondent's company size, gender, education, and job title. In order to test some hypothesis in this context, we have formulated the following null hypotheses:

H₁—The perception of the users regarding the impact of CIM on employment does not depend on the gender.

H₂—The perception of the users regarding the stress to be placed on CIM initiatives does not depend on the gender.

H₃—The perception of the users regarding the impact of CIM on employment does not depend on their company's size.

H₄—The perception of the users regarding the stress to be placed on CIM initiatives does not depend on their company's size.

Table 1 One-way ANOVA: Perception of Various Groups of Employees toward the Impact of CIM on Employment and Stress on CIM

By gender		Sum of squares	d.f.	Mean square	F	Sig.
CIM-IMP	Between groups	1.435	1	1.435	2.669	0.104
	Within groups	112.394	209	0.538		
	Total	113.829	210			
Stress	Between groups	.557	1	0.557	0.465	0.496
	Within groups	249.974	209	1.196		
	Total	250.531	210			
By the size of the companies		Sum of squares	d.f.	Mean square	F	Sig.
CIM-IMP	Between groups	1.254	3	0.418	0.769	0.513
	Within groups	112.575	207	0.544		
	Total	113.829	210			
Stress	Between groups	1.904	3	0.635	0.528	0.663
	Within groups	248.627	207	1.201		
	Total	250.531	210			
By education		Sum of squares	d.f.	Mean square	F	Sig.
CIM-IMP	Between groups	1.904	6	0.317	0.578	0.747
	Within groups	111.925	204	0.549		
	Total	113.829	210			
Stress	Between groups	13.809	6	2.301	1.983	0.070
	Within groups	236.722	204	1.160		
	Total	250.531	210			
By the title of respondents		Sum of squares	d.f.	Mean square	F	Sig.
CIM-IMP	Between groups	1.580	3	0.527	0.971	0.407
	Within groups	112.250	207	0.542		
	Total	113.829	210			
Stress	Between groups	62.563	3	20.854	22.966	0.000
	Within groups	187.967	207	0.908		
	Total	250.531	210			

H₅—The perception of the users regarding the impact of CIM on employment does not depend on the managerial and functional title of the respondents.

H₆—The perception of the users regarding the stress to be placed on CIM initiatives does not depend on the managerial and functional title of the respondents.

H₇—The perception of the users regarding the impact of CIM on employment does not depend on the education of the user.

H₈—The perception of the users regarding the stress to be placed on CIM initiatives does not depend on the user's education.

These hypotheses can be formulated as follows. We use $AP_i(X)$ to represent the average perception of group i of respondents toward the specific dimension of CIM

that is under examination, where for gender: $i = M$ (male), and $i = F$ (female); for size: $i = s$ (small), m (mid-size), and l (large); for education: $i = 1$ (high school or less), 2 (some college), 3 (bachelor's degree), 4 (some graduate study), 5 (graduate degree), and 6 (professional training); and for job title dimension: $i = a$ (CIM managers), b (IS professionals), c (CIM professionals), and d (clerical and operating personnel). For instance, $AP_M(\text{CIM-IMP})$ denotes the average perception of males toward the CIM impact on employment, $AP_M(\text{Stress})$ represents the average perception of males toward stress to be placed on CIM initiatives and management, and $AP_b(\text{Stress})$ denotes the average perception of IS professionals toward stress to be placed on CIM initiatives and management.

Using these definitions, we can formulate the above hypotheses, respectively, in the following mathematical forms. For the gender factor:

$$H_1: AP_M(\text{CIM-IMP}) = AP_F(\text{CIM-IMP})$$

$$H_2: AP_M(\text{Stress}) = AP_F(\text{Stress})$$

For the company size factor:

$$H_3: AP_s(\text{CIM-IMP}) = AP_m(\text{CIM-IMP}) = AP_1(\text{CIM-IMP})$$

$$H_4: AP_s(\text{Stress}) = AP_m(\text{Stress}) = AP_1(\text{Stress})$$

For the education factor:

$$H_5: AP_1(\text{CIM-IMP}) = AP_2(\text{CIM-IMP}) = AP_3(\text{CIM-IMP}) = AP_4(\text{CIM-IMP}) = AP_5(\text{CIM-IMP}) = AP_6(\text{CIM-IMP})$$

$$H_6: AP_1(\text{Stress}) = AP_2(\text{Stress}) = AP_3(\text{Stress}) = AP_4(\text{Stress}) = AP_5(\text{Stress}) = AP_6(\text{Stress})$$

For the job responsibility and title factor:

$$H_7: AP_a(\text{CIM-IMP}) = AP_b(\text{CIM-IMP}) = AP_c(\text{CIM-IMP}) = AP_d(\text{CIM-IMP})$$

$$H_8: AP_a(\text{Stress}) = AP_b(\text{Stress}) = AP_c(\text{Stress}) = AP_d(\text{Stress})$$

One-way ANOVA, as shown in Table I, suggests that the first three null hypotheses can be accepted, i.e., there were no significant differences among various groups of respondents by the size of companies, by the gender, and by education of respondents, as they perceive the impact of CIM on employment and the stress to be placed on CIM. However, the data reported two different results for the fourth null hy-

pothesis (H_4): While CIM management, CIM and IS professionals, and operating personnel have the same perception toward the impact of CIM, they show significantly different perceptions toward the stress to be placed on CIM initiatives and management.

The result of ANOVA rejects the fifth hypothesis (H_5) that the various employees have similar perceptions of stress on CIM. We extend our analysis to pinpoint whether the observed differences in the survey could be attributed to just the natural variability among the sample averages or whether there is reason to believe that some of the four groups have significantly different perceptions for the stress to be placed on CIM over the next two years. By extending our analysis to a multiple comparison procedure, using Bonferroni, as shown in Table II, one can see no significant difference between the perception of CIM managers and CIM professionals on the one hand and between IS professionals and clerical/operating personnel on the other hand. However, the first two groups, CIM managers and CIM professionals, have significantly different perceptions regarding the stress to be placed on CIM initiatives and management. In particular, IS professionals show significantly lower stress on CIM projects.

We further expanded our analysis of the data to include more than one grouping or factor variable. In particular, by reviewing the one-factor analysis, we combined employment title and the educational background of the respondents in our two-way analysis, as shown in Table III. There was no significant evidence to conclude that there exists interaction between the job

Table II Multiple Comparisons (Bonferroni) of Respondents' Perceptions by the Employment Position

Dependent Variable: STRESS						
By the employment position					95% Confidence interval	
(I) TITLE	(J) TITLE	Mean difference (I-J)	Std. error	Sig.	Lower bound	Upper bound
CIM managers	IS professionals	1.17 ^a	0.19	0.000	0.67	1.68
	CIM professionals	-8.67E-02	0.18	1.000	-0.57	0.40
	Clerical and operation	0.88 ^a	0.19	0.000	0.39	1.38
IS professionals	CIM managers	-1.17 ^a	0.19	0.000	-1.68	-0.67
	CIM professionals	-1.26 ^a	0.19	0.000	-1.76	-0.77
	Clerical and operation	-0.29	0.19	0.749	-0.80	0.21
CIM professionals	CIM managers	8.67E-02	0.18	1.000	-0.40	0.57
	IS professionals	1.26 ^a	0.19	0.000	0.77	1.76
	Clerical and operation	0.97 ^a	0.18	0.000	0.48	1.46
Clerical and operation	CIM managers	-0.88 ^a	0.19	0.000	-1.38	-0.39
	IS professionals	0.29	0.19	0.749	-0.21	0.80
	CIM professionals	-0.97 ^a	0.18	0.000	-1.46	-0.48

^a The mean difference is significant at the 0.05 level.

Table III Tests of Interaction between Factors: Impact of CIM and Stress on CIM**Dependent Variable: CIM Impact on the Number of Employees****Job title & the company size**

Source	Type III sum of squares	d.f.	Mean square	F	Sig.
Corrected model	9.226	10	0.923	1.764	0.069
Intercept	56.705	1	56.705	108.420	0.000
Title	6.874	3	2.291	4.381	0.005
Size	1.432	3	0.477	0.913	0.436
Title * Size	6.606	4	1.652	3.158	0.015
Error	104.603	200	0.523		
Total	934.000	211			
Corrected total	113.829	210	Squared = 0.081 (adjusted R squared = 0.035)		

Job title and gender	Type III sum of squares	d.f.	Mean square	F	Sig.
Corrected Model	5.642	7	0.806	1.512	0.165
Intercept	599.524	1	599.524	1124.932	0.000
Title	1.682	3	0.561	1.052	0.371
Gender	0.879	1	0.879	1.650	0.200
Title * Gender	2.893	3	0.964	1.809	0.147
Error	108.187	203	0.533		
Total	934.000	211			
Corrected total	113.829	210	a. R squared = 0.050 (adjusted R squared = 0.017)		

Job title and education	Type III sum of squares	d.f.	Mean square	F	Sig.
Corrected model	16.658	22	0.757	1.465	0.090
Intercept	232.932	1	232.932	450.660	0.000
Title	1.943	3	0.648	1.253	0.292
Education	2.564	6	0.427	0.827	0.551
Title * Education	13.277	13	1.021	1.976	0.025
Error	97.171	188	0.517		
Total	934.000	211			
Corrected total	113.829	210	a. R squared = 0.146 (adjusted R squared = 0.046)		

Dependent Variable: STRESS**Job title and company size**

Job title and company size	Type III sum of squares	d.f.	Mean square	F	Sig.
Corrected Model	68.092	10	6.809	7.465	0.000
Intercept	226.731	1	226.731	248.556	0.000
Title	43.736	3	14.579	15.982	0.000
Size	1.018	3	0.339	0.372	0.773
Title * Size	5.076	4	1.269	1.391	0.238
Error	182.439	200	0.912		
Total	2442.000	211			
Corrected total	250.531	210	a. R squared = 0.272 (adjusted R squared = 0.235)		

Job title and gender	Type III sum of squares	d.f.	Mean square	F	Sig.
Corrected Model	64.646	7	9.235	10.085	0.000
Intercept	1542.226	1	1542.226	1684.226	0.000
Title	48.096	3	16.032	17.508	0.000
Gender	0.421	1	0.421	0.460	0.499
Title * Gender	1.665	3	0.555	0.606	0.612
Error	185.885	203	0.916		
Total	2442.000	211			
Corrected total	250.531	210	a. R squared = 0.258 (adjusted R squared = 0.232)		

(continues)

Table III (continued)

Job title and education	Type III sum of squares	d.f.	Mean square	F	Sig.
Corrected Model	95.594	22	4.345	5.272	0.000
Intercept	793.582	1	793.582	962.930	0.000
Title	11.842	3	3.947	4.790	0.003
Education	2.872	6	0.479	0.581	0.745
Title * Education	25.306	13	1.947	2.362	0.006
Error	154.937	188	0.824		
Total	2442.000	211			
Corrected total	250.531	210	a. R squared = 0.382 (adjusted R squared = 0.309)		

title of respondents and their gender, and the size of their companies. Therefore, one cannot reject the sixth null hypothesis (H_6), that there is no interaction between the employment title and the gender as well as between the employment title and the size of the company with regard to stress on CIM. In other words, the effect of job title and responsibility on placing stress on CIM seem to be similar for males and females and for respondents from mid-size and large companies. However, there was a significant interaction between the job title and the education of employees. The significant interaction between employment title and educational background of respondents tells us that it is reasonable to believe that the difference in perception among respondents with different education is significantly different for any specific job title of the employees. Thus the relationship between job title and perception of employees is different with educational background, i.e., the eighth null hypothesis (H_8) is not supported.

VII. CONCLUDING COMMENTS

CIM in this study was considered as the (vertical) integration of the flow of information from factory to the board room with its focus on business and also it is the (horizontal) integration of functional areas from design and manufacturing, to marketing and other business functions. It is, thus, a strategic challenge for senior management, particularly for CIOs, to integrate CIM strategy into corporate strategy to ensure the company's competitive effectiveness.

Our empirical study in this investigation was conducted to analyze the perception of various users, including IS professionals and CIM managers, with regard to the impact of CIM on employment and the stress to be placed on CIM projects. According to the results of this empirical study, summarized in Table IV, two types of perceptions were recognized: one perception, shared by CIM managers and professionals,

Table IV Summary Results of the Statistical Analysis

Null hypothesis	Statement of the hypothesis	Empirical result
H ₁	The perception of the users regarding the impact of CIM on employment does not depend on the gender	Supported
H ₂	The perception of the users regarding the stress to be placed on CIM initiatives does not depend on the gender	Supported
H ₃	The perception of the users regarding the impact of CIM on employment does not depend on their company's size	Supported
H ₄	The perception of the users regarding the stress to be placed on CIM initiatives does not depend on their company's size	Not supported
H ₅	The perception of the users regarding the impact of CIM on employment does not depend on the managerial and functional title of the respondents	Not supported
H ₆	The perception of the users regarding the stress to be placed on CIM initiatives does not depend on the managerial and functional title of the respondents	Supported
H ₇	The perception of the users regarding the impact of CIM on employment does not depend on the education of the user	Not supported
H ₈	The perception of the users regarding the stress to be placed on CIM initiatives does not depend on the user's education	Not supported

judged the impact of CIM insignificant, while another, expressed by IS professionals and clerical operators, viewed the impact of CIM as labor replacement—i.e., decreasing the number of employees. The latter judgement was also displayed by the majority of employees from the mid-size companies.

According to the data, the perceived stress on CIM did not depend on the organizational size, but rather upon the type of responding employee. Two groups with two different perceptions were identified: CIM managers and professionals viewed that there would be significant stress on CIM, whereas IS professionals and clerical operators were not as optimistic. Our analysis suggests that IS professionals perceive the stress on CIM to be significantly different from CIM managers and CIM professionals. In particular, IS professionals have placed less emphasis on CIM undertakings and their management, and thus lagged behind other departments in contributing to the CIM development and its strategic applications for competitive edge. These data also suggest that training and education among CIM managers and CIM professionals has played a positive role in placing higher stress on CIM technology.

Successful implementation and management of CIM initiative depend not only upon expertise in technology assessment and deployment, but more importantly, on the business vision and a company-wide management approach. Effective assessment in deployment of CIM technology, in particular, should take into consideration its potential values and non-tangible as well as tangible costs and benefits. Given the scope of CIM technology, senior IS management with its long-range and company-wide view is responsible for playing a leading role in CIM undertakings, for estimating all possible strategic and intangible benefits resulting from CIM, and for leading the feasibility analysis and the adoption process. The lack of leadership by senior IS management in CIM initiatives and management will inevitably lead to the automation of functional tasks without achieving the integration that is vital to the survival and functioning of the company as a whole.

SEE ALSO THE FOLLOWING ARTICLES

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